

Detection of Excess Returns in AI-Related Stocks

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Abstract

This project examines whether AI-related stocks experienced statistically significant positive excess returns after the release of ChatGPT on November 30, 2022. Daily stock price data for the Magnificent Seven companies and SPY were collected from Yahoo Finance using the `tidyquant` package in R. Adjusted closing prices were used to calculate daily log returns, and excess returns were calculated by subtracting SPY returns from each stock's return. The analysis focused on a 30-calendar-day window before and after the ChatGPT release. Visualizations, descriptive statistics, paired t-tests, and linear regression models were used to compare pre-event and post-event performance. The results did not provide strong statistical evidence that AI-related stocks had significantly higher excess returns after the ChatGPT release. The paired t-tests were not statistically significant, and the regression models showed that SPY returns were significant predictors, while the post-event indicator was not significant. Overall, the findings suggest that market movement explained stock returns more strongly than the ChatGPT event during this window.

Introduction

Recent advances in artificial intelligence, especially the release of ChatGPT, have increased investor attention toward major technology firms and the possibility of an “AI bubble.” Companies such as NVIDIA, Microsoft, Meta, Alphabet, Amazon, Apple, and Tesla are often connected to AI growth because of their roles in chips, cloud computing, software, platforms, and consumer technology.

This project examines whether AI-related assets showed statistically significant positive excess returns after the release of ChatGPT on November 30, 2022. In this project, “AI-related asset behavior” refers to changes in daily returns and excess returns before and after the ChatGPT release.

The main question of research is:

Do AI-related assets experience significant positive excess returns relative to the market after the release of ChatGPT?

To answer this question, this project will:

- Analyze daily returns and excess returns of major AI-related assets from 2020 to the present.
- Compare excess returns before and after the ChatGPT release.
- Evaluate performance relative to SPY.
- Test whether post-event excess returns are statistically greater than pre-event excess returns.

The hypotheses for the paired t-test are:

$$H_0: \mu_d = 0$$

$$H_1: \mu_d > 0$$

where d = post-event excess return – pre-event excess return.

Data Collection and Description

Daily stock price data will be collected from Yahoo Finance using the `tidyquant` package in R. The analysis will use adjusted closing prices to calculate daily log returns, market returns, excess returns, and a post-event indicator.

AI-related assets: The Magnificent Seven

Ticker	Company	AI Category
AAPL	Apple	AI Devices / Consumer Technology
AMZN	Amazon	AI Cloud / E-commerce
GOOGL	Alphabet	AI Search / Cloud
TSLA	Tesla	AI / Autonomous Driving
META	Meta	AI Platforms / Advertising
MSFT	Microsoft	AI Software / Cloud
NVDA	Nvidia	AI Chips / Hardware

Benchmark:

Ticker	Description
SPY	S&P 500 ETF used as the market benchmark

Key event:

- November 30, 2022: Release of ChatGPT

Relevant variables:

Variable	Type	Description
----------	------	-------------

Variable	Type	Description
Date	Date	Trading day for each stock or benchmark
Symbol	Categorical	Stock ticker, such as NVDA, MSFT, or SPY
Adjusted closing price	Numeric	Closing price adjusted for stock splits and dividends
Daily return	Numeric	Log return calculated from adjusted closing prices
Market return	Numeric	Daily log return of SPY
Excess return	Numeric	Stock return minus SPY return
Post-event indicator	Binary	Equals 1 after the ChatGPT release and 0 before the release

Methodology

The analysis will focus on a 30-calendar-day window before and after the ChatGPT release on November 30, 2022. Visualizations will be used to examine stock price trends and excess return patterns. Descriptive statistics such as mean and standard deviation will be compared between pre-event and post-event excess returns.

Excess returns will be calculated as:

$$\text{Excess Return} = \text{Stock Return} - \text{SPY Return}$$

A paired t-test will be used to test whether average post-event excess returns are greater than average pre-event excess returns:

$$H_0: \mu_d = 0$$

$$H_1: \mu_d > 0$$

where d = post-event excess return – pre-event excess return.

A linear regression model will also be used to control market returns:

$$\text{Stock Return} = \beta_0 + \beta_1 \text{SPY Return} + \beta_2 \text{PostEvent} + \epsilon$$

In this model, β_2 measures whether stock returns were higher after the ChatGPT release, controlling overall market performance.

Return and Excess Return Calculation

Daily log returns are calculated using adjusted closing prices:

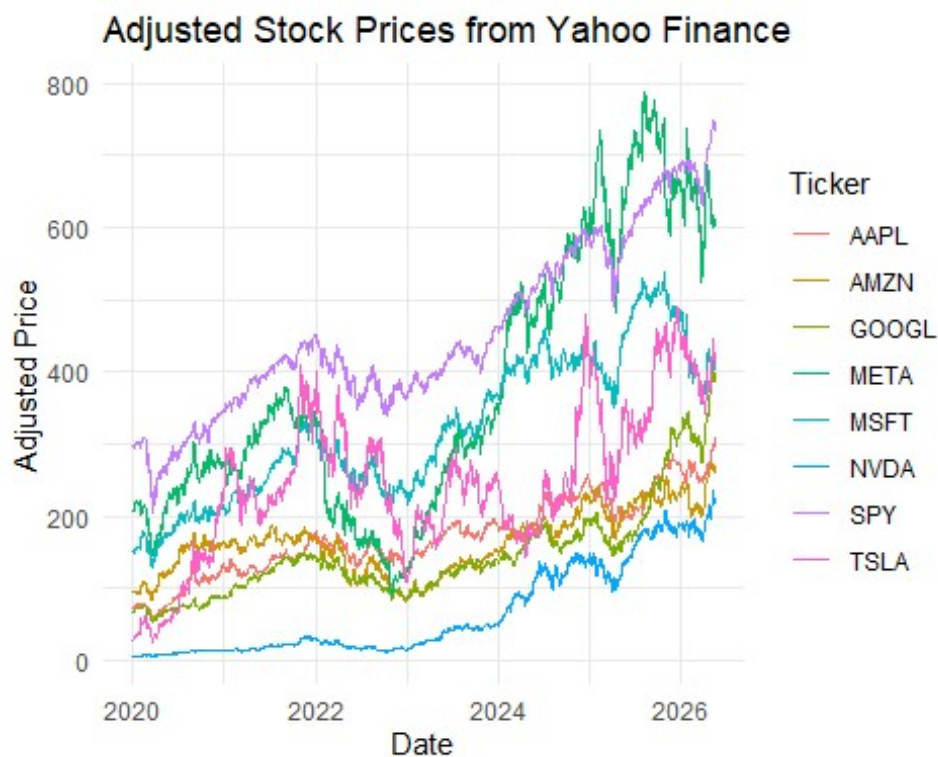
$$\text{Daily Log Return} = \log\left(\frac{\text{Adjusted Price}_t}{\text{Adjusted Price}_{t-1}}\right)$$

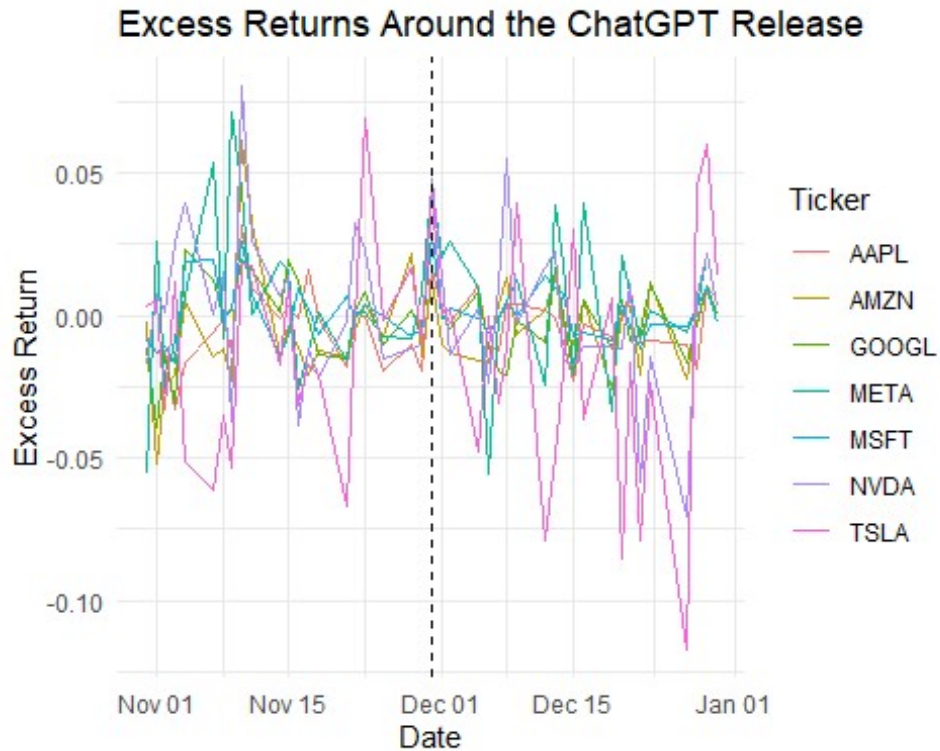
Excess returns are calculated as:

$$\text{Excess Return} = \text{Log Return} - \text{SPY Return}$$

Analysis and Results

Data Visualization





Descriptive Statistics

In the 30-calendar-day window around the ChatGPT release, the average log return across the selected stocks was negative, around -0.0027, and the average excess return was also slightly negative, around -0.00245. This suggests that, overall, the selected AI-related stocks did not strongly outperform SPY during this event window.

Paired T-Test

The paired t-test results do not provide statistically significant evidence that AI-related stocks had higher excess returns after the ChatGPT release. Although some stocks had positive average differences between post-event and pre-event excess returns, none of these differences were significant at the 5% level. Therefore, the paired t-tests do not support the alternative hypothesis.

Linear Regression

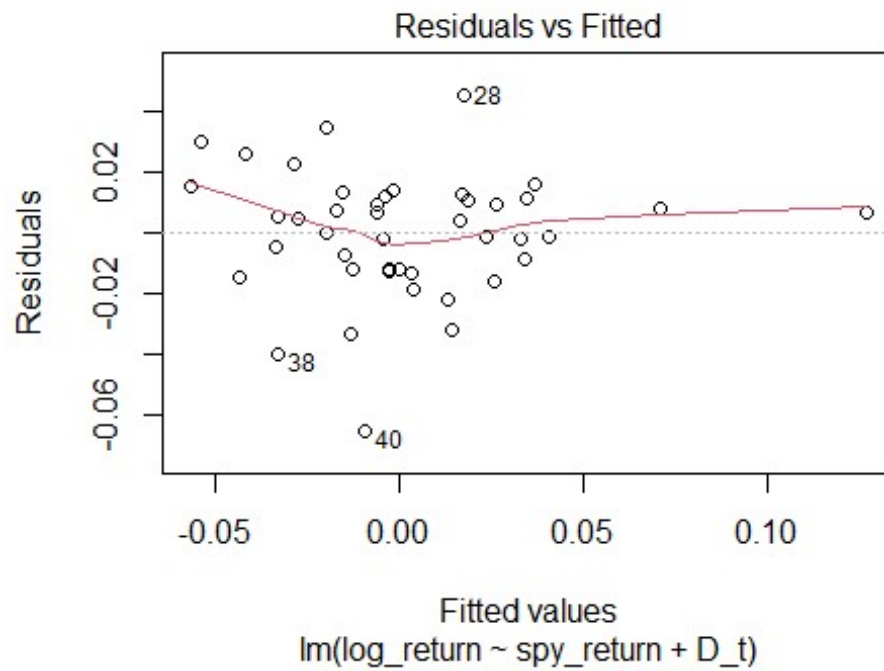
The regression model is:

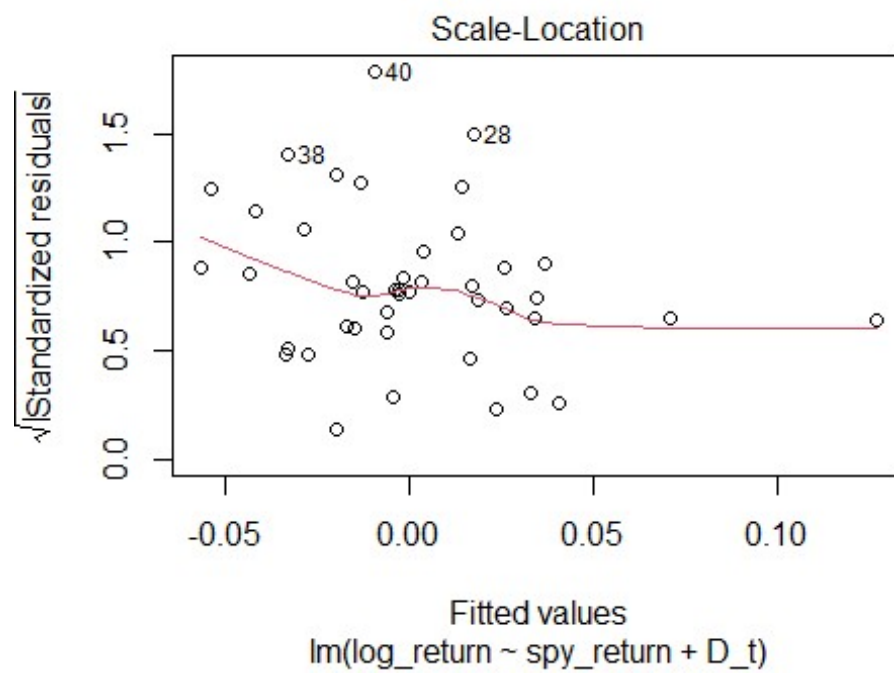
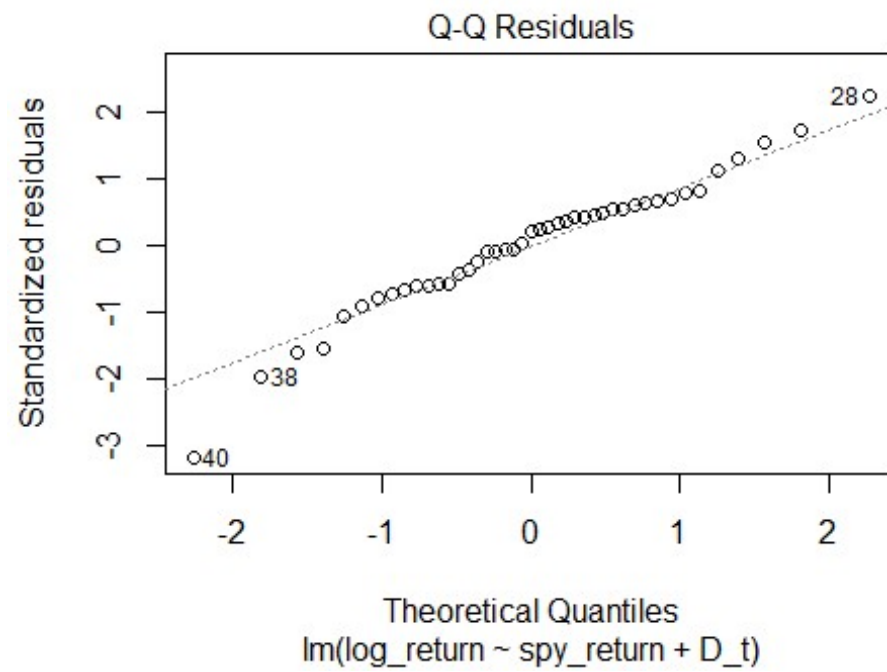
$$\text{Stock Return} = \beta_0 + \beta_1 \text{SPY Return} + \beta_2 \text{PostEvent} + \epsilon$$

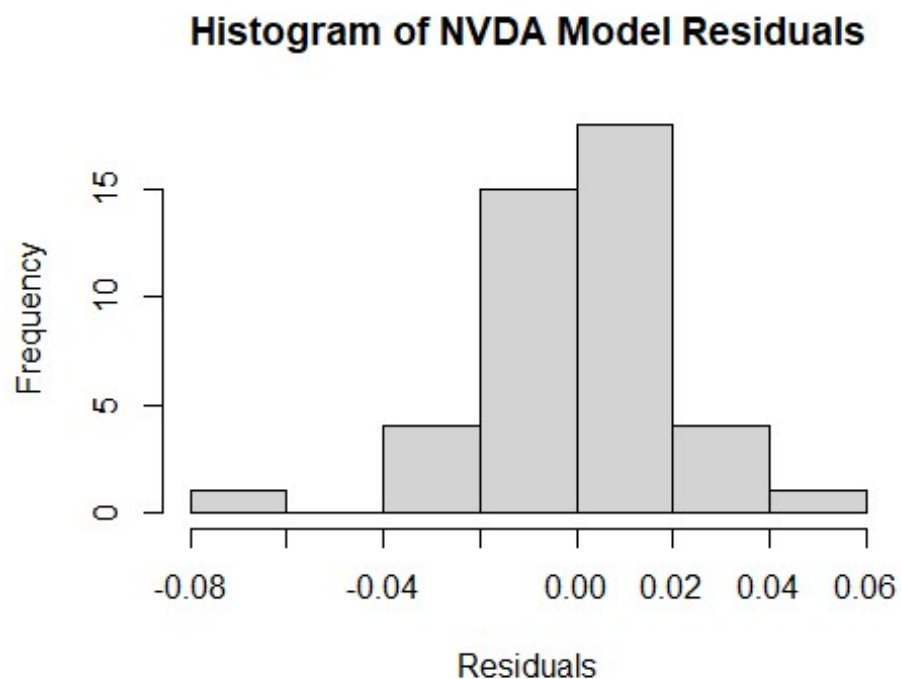
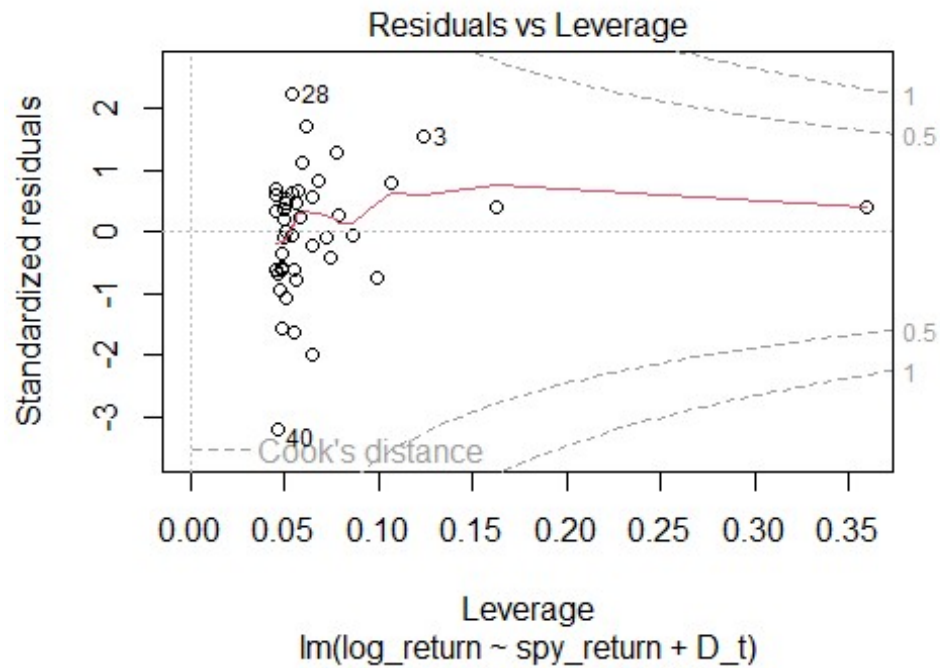
The regression results show that SPY return was statistically significant for all stocks, indicating that overall market movement was an important predictor of each stock's return. However, the post-event indicator was not statistically significant for any stock. This

suggests that after controlling for SPY, the ChatGPT release did not produce a statistically significant increase in stock returns within the 30-calendar-day event window.

Model Diagnostics







Model diagnostics were checked using NVDA as an example because NVIDIA is one of the most directly AI-related companies in the sample. The diagnostic plots help evaluate whether regression assumptions are reasonable. The residual histogram and regression

diagnostic plots suggest that the model may have some deviations from normality and possible outliers, which is expected in short-window financial return data. This is a limitation of the analysis, but the regression still provides useful evidence about the relationship between NVDA returns, SPY returns, and the post-event period.

Conclusion

Overall, the results do not provide strong statistical evidence that the ChatGPT release caused significant positive excess returns for the selected AI-related stocks. The paired t-tests found no significant post-event increase in excess returns, and the regression models showed that the post-event indicator was not significant after controlling SPY. However, SPY return was significant across all models, showing that general market movement was an important driver of stock returns during the event window. Therefore, the hypothesis is not supported by the results in this analysis. A limitation is that the study focuses on one event and a short event window, so future research could compare other AI events such as DeepSeek or Claude Mythos separately.

References

Business Science. (n.d.). *tidyquant: Tidy quantitative financial analysis*. <https://business-science.github.io/tidyquant/>

Yahoo Finance. (n.d.). *Historical stock price data*. <https://finance.yahoo.com/>

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R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). *dplyr: A grammar of data manipulation*. <https://dplyr.tidyverse.org/>

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York.

Appendix

R Code

```
tickers <- c("AAPL", "AMZN", "GOOGL", "TSLA", "META", "MSFT", "NVDA", "SPY")

prices <- tq_get(
  tickers,
  from = "2020-01-01",
  to = Sys.Date(),
  get = "stock.prices"
)

head(prices)

## # A tibble: 6 × 8
##   symbol date      open  high  low close  volume adjusted
##   <chr>  <date>    <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>
## 1 AAPL   2020-01-02  74.1  75.2  73.8  75.1  135480400    72.3
## 2 AAPL   2020-01-03  74.3  75.1  74.1  74.4  146322800    71.6
## 3 AAPL   2020-01-06  73.4  75.0  73.2  74.9  118387200    72.2
## 4 AAPL   2020-01-07  75.0  75.2  74.4  74.6  108872000    71.9
## 5 AAPL   2020-01-08  74.3  76.1  74.3  75.8  132079200    73.0
## 6 AAPL   2020-01-09  76.8  77.6  76.6  77.4  170108400    74.6

tail(prices)

## # A tibble: 6 × 8
##   symbol date      open  high  low close  volume adjusted
```

```
##   <chr>   <date>      <dbl> <dbl> <dbl> <dbl>      <dbl>      <dbl>
## 1 SPY    2026-05-15   742.  743.  738.  739. 60410800    739.
## 2 SPY    2026-05-18   740.  741.  733.  739. 47843900    739.
## 3 SPY    2026-05-19   735.  738.  732.  734. 54255900    734.
## 4 SPY    2026-05-20   736.  742.  734.  741. 45768000    741.
## 5 SPY    2026-05-21   739.  745.  737.  743. 43332200    743.
## 6 SPY    2026-05-22   746.  749.  744.  746. 41671800    746.
```

```
returns <- prices %>%
  group_by(symbol) %>%
  arrange(date) %>%
  mutate(log_return = log(adjusted / lag(adjusted))) %>%
  ungroup() %>%
  na.omit()
```

```
head(returns)
```

```
## # A tibble: 6 × 9
##   symbol date      open  high  low close  volume adjusted log_return
##   <chr>   <date>    <dbl> <dbl> <dbl> <dbl>      <dbl>      <dbl>
## 1 AAPL   2020-01-03   74.3  75.1  74.1  74.4 146322800    71.6   -0.00977
## 2 AMZN   2020-01-03   93.2  94.3  93.2  93.7  75288000    93.7   -0.0122
## 3 GOOGL  2020-01-03   67.4  68.7  67.4  68.1  23408000    67.5   -0.00524
## 4 TSLA   2020-01-03   29.4  30.3  29.1  29.5 266677500    29.5    0.0292
## 5 META   2020-01-03  207.  210.  207.  209.  11188400   207.   -0.00531
## 6 MSFT   2020-01-03  158.  160.  158.  159.  21116200   150.   -0.0125
```

```
SPY_returns <- returns %>%
  filter(symbol == "SPY") %>%
  select(date, spy_return = log_return)
```

```
head(SPY_returns)
```

```
## # A tibble: 6 × 2
##   date      spy_return
##   <date>      <dbl>
## 1 2020-01-03   -0.00760
## 2 2020-01-06    0.00381
## 3 2020-01-07   -0.00282
## 4 2020-01-08    0.00532
## 5 2020-01-09    0.00676
## 6 2020-01-10   -0.00288
```

```
excess_returns <- returns %>%
  filter(symbol != "SPY") %>%
  left_join(SPY_returns, by = "date") %>%
  mutate(excess_return = log_return - spy_return)
```

```
head(excess_returns)
```

```
## # A tibble: 6 × 11
##   symbol date       open  high  low close  volume adjusted log_return
##   <chr>  <date>      <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>    <dbl>
## 1 AAPL   2020-01-03   74.3  75.1  74.1  74.4 146322800    71.6   -0.00977
## 2 AMZN   2020-01-03   93.2  94.3  93.2  93.7  75288000    93.7   -0.0122
## 3 GOOGL  2020-01-03   67.4  68.7  67.4  68.1  23408000    67.5   -0.00524
## 4 TSLA   2020-01-03   29.4  30.3  29.1  29.5  266677500    29.5    0.0292
## 5 META   2020-01-03  207.  210.  207.  209.  11188400    207.   -0.00531
## 6 MSFT   2020-01-03  158.  160.  158.  159.  21116200    150.   -0.0125
## # i 2 more variables: spy_return <dbl>, excess_return <dbl>

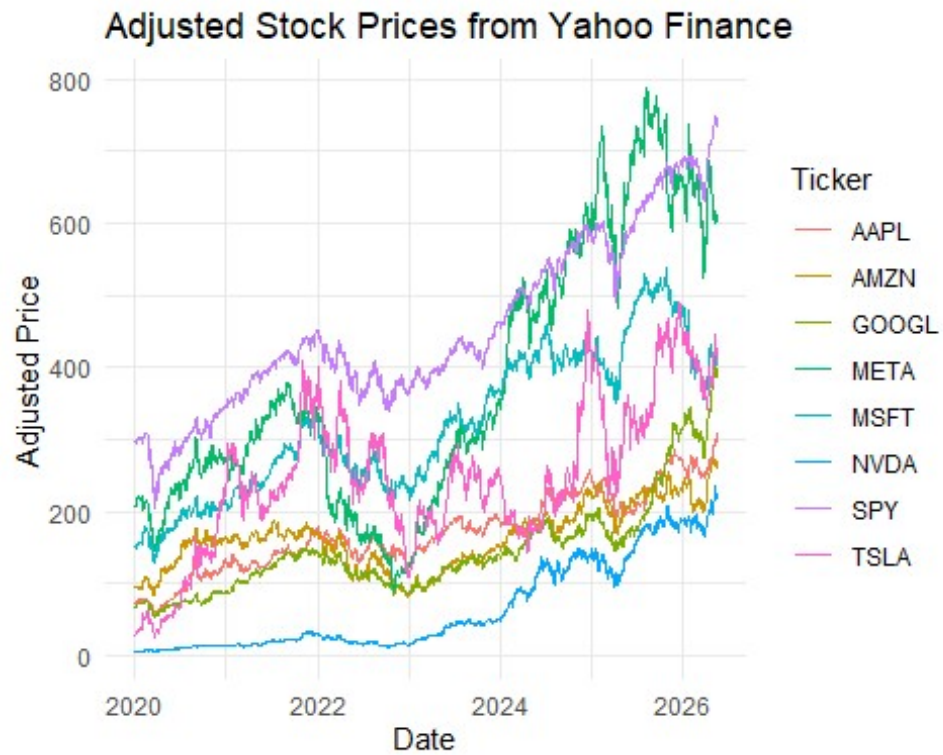
releaseChatGPT <- as.Date("2022-11-30")

event_window <- excess_returns %>%
  filter(date >= releaseChatGPT - 30,
         date <= releaseChatGPT + 30) %>%
  mutate(
    D_t = ifelse(date >= releaseChatGPT, 1, 0),
    period = ifelse(D_t == 1, "Post-event", "Pre-event")
  )

event_window %>%
  select(symbol, date, log_return, spy_return, excess_return, D_t, period) %>%
  arrange(symbol, date)

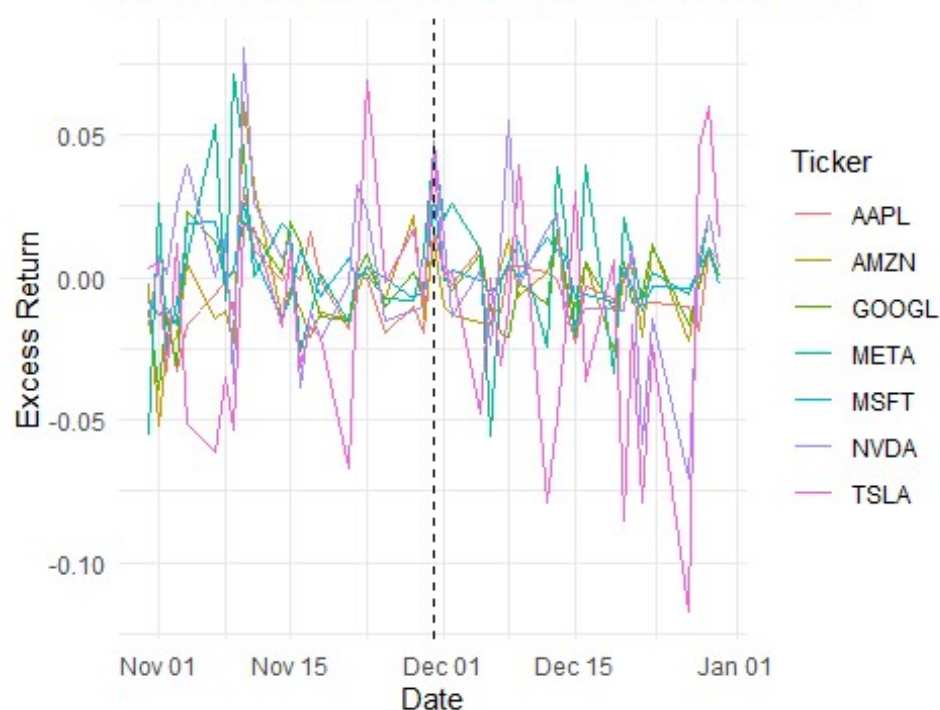
## # A tibble: 301 × 7
##   symbol date       log_return spy_return excess_return  D_t period
##   <chr>  <date>      <dbl>    <dbl>    <dbl> <dbl> <chr>
## 1 AAPL   2022-10-31   -0.0155   -0.00725   -0.00828    0 Pre-event
## 2 AAPL   2022-11-01   -0.0177   -0.00439   -0.0133    0 Pre-event
## 3 AAPL   2022-11-02   -0.0380   -0.0254   -0.0126    0 Pre-event
## 4 AAPL   2022-11-03   -0.0433   -0.0104   -0.0330    0 Pre-event
## 5 AAPL   2022-11-04   -0.00195    0.0143   -0.0162    0 Pre-event
## 6 AAPL   2022-11-07    0.00389    0.00952   -0.00563    0 Pre-event
## 7 AAPL   2022-11-08    0.00417    0.00538   -0.00121    0 Pre-event
## 8 AAPL   2022-11-09   -0.0338   -0.0208   -0.0129    0 Pre-event
## 9 AAPL   2022-11-10    0.0852    0.0535    0.0317    0 Pre-event
## 10 AAPL  2022-11-11    0.0191    0.00963    0.00945    0 Pre-event
## # i 291 more rows

ggplot(prices, aes(x = date, y = adjusted, color = symbol)) +
  geom_line() +
  labs(
    title = "Adjusted Stock Prices from Yahoo Finance",
    x = "Date",
    y = "Adjusted Price",
    color = "Ticker"
  ) +
  theme_minimal()
```



```
ggplot(event_window, aes(x = date, y = excess_return, color = symbol)) +
  geom_line() +
  geom_vline(xintercept = releaseChatGPT, linetype = "dashed") +
  labs(
    title = "Excess Returns Around the ChatGPT Release",
    x = "Date",
    y = "Excess Return",
    color = "Ticker"
  ) +
  theme_minimal()
```

Excess Returns Around the ChatGPT Release



```
summary(event_window)
```

```
##      symbol              date              open              high
## Length:301             Min.   :2022-10-31   Min.    : 13.04   Min.    : 13.78
## Class :character       1st Qu.:2022-11-14   1st Qu.: 90.50   1st Qu.: 91.18
## Mode  :character       Median :2022-11-30   Median :113.76   Median :116.27
##                               Mean   :2022-11-29   Mean    :124.57   Mean    :126.45
##                               3rd Qu.:2022-12-15   3rd Qu.:159.25   3rd Qu.:160.99
##                               Max.    :2022-12-30   Max.    :261.69   Max.    :263.92
##      low              close              volume              adjusted
## Min.   : 12.96   Min.   : 13.22   Min.   : 9200800   Min.   : 13.20
## 1st Qu.: 88.23   1st Qu.: 89.98   1st Qu.: 30665100   1st Qu.: 89.52
## Median :111.02   Median :113.93   Median : 68086900   Median :113.04
## Mean   :122.10   Mean   :124.08   Mean   :121364728   Mean   :122.52
## 3rd Qu.:153.28   3rd Qu.:156.80   3rd Qu.:109536700   3rd Qu.:156.80
## Max.   :254.31   Max.   :257.22   Max.   :710894000   Max.   :250.10
##      log_return      spy_return      excess_return      D_t
## Min.   :-0.121138   Min.   :-0.0254164   Min.   :-0.117187   Min.   :0.00
## 00
## 1st Qu.: -0.019788   1st Qu.: -0.0085432   1st Qu.: -0.012936   1st Qu.:0.00
## 00
## Median :-0.004950   Median :-0.0017191   Median :-0.001215   Median :1.00
## 00
## Mean   :-0.002741   Mean   :-0.0002908   Mean   :-0.002450   Mean   :0.51
## 16
## 3rd Qu.: 0.012323   3rd Qu.: 0.0078034   3rd Qu.: 0.009415   3rd Qu.:1.00
```

```

00
## Max. : 0.133912 Max. : 0.0534975 Max. : 0.080415 Max. : 1.00
00
## period
## Length:301
## Class :character
## Mode :character
##
##
##

window_aapl <- event_window %>% filter(symbol == "AAPL")
window_amzn <- event_window %>% filter(symbol == "AMZN")
window_googl <- event_window %>% filter(symbol == "GOOGL")
window_meta <- event_window %>% filter(symbol == "META")
window_msft <- event_window %>% filter(symbol == "MSFT")
window_nvda <- event_window %>% filter(symbol == "NVDA")
window_tsla <- event_window %>% filter(symbol == "TSLA")

head(window_aapl)

## # A tibble: 6 × 13
##   symbol date      open high low close volume adjusted log_return
##   <chr> <date>    <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl>    <dbl>
## 1 AAPL  2022-10-31  153.  154.  152.  153.  97943200    151.   -0.0155
## 2 AAPL  2022-11-01  155.  155.  149.  151.  80379300    148.   -0.0177
## 3 AAPL  2022-11-02  149.  152.  145.  145.  93604600    142.   -0.0380
## 4 AAPL  2022-11-03  142.  143.  139.  139.  97918500    136.   -0.0433
## 5 AAPL  2022-11-04  142.  143.  134.  138. 140814800    136.   -0.00195
## 6 AAPL  2022-11-07  137.  139.  136.  139.  83374600    137.    0.00389
## # i 4 more variables: spy_return <dbl>, excess_return <dbl>, D_t <dbl>,
## #   period <chr>

paired_test_stock <- function(stock_window) {
  before <- stock_window$excess_return[stock_window$D_t == 0]
  after <- stock_window$excess_return[stock_window$D_t == 1]

  n <- min(length(before), length(after))

  t.test(after[1:n], before[1:n],
         paired = TRUE,
         alternative = "greater")
}

paired_test_stock(window_aapl)

##
## Paired t-test
##
## data: after[1:n] and before[1:n]

```

```
## t = 0.59906, df = 20, p-value = 0.2779
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -0.004712905      Inf
## sample estimates:
## mean difference
##      0.00250815
```

`paired_test_stock(window_amzn)`

```
##
## Paired t-test
##
## data: after[1:n] and before[1:n]
## t = 0.58456, df = 20, p-value = 0.2827
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -0.005620271      Inf
## sample estimates:
## mean difference
##      0.002881548
```

`paired_test_stock(window_googl)`

```
##
## Paired t-test
##
## data: after[1:n] and before[1:n]
## t = -0.19127, df = 20, p-value = 0.5749
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -0.009734726      Inf
## sample estimates:
## mean difference
##      -0.0009718055
```

`paired_test_stock(window_meta)`

```
##
## Paired t-test
##
## data: after[1:n] and before[1:n]
## t = 0.18879, df = 20, p-value = 0.4261
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -0.01441754      Inf
## sample estimates:
## mean difference
##      0.001772173
```

`paired_test_stock(window_msft)`

```
##
## Paired t-test
##
## data: after[1:n] and before[1:n]
## t = 0.3259, df = 20, p-value = 0.3739
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -0.004663941      Inf
## sample estimates:
## mean difference
##      0.001086626

paired_test_stock(window_nvda)

##
## Paired t-test
##
## data: after[1:n] and before[1:n]
## t = -0.81997, df = 20, p-value = 0.789
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -0.02207604      Inf
## sample estimates:
## mean difference
##      -0.007113493

paired_test_stock(window_tsla)

##
## Paired t-test
##
## data: after[1:n] and before[1:n]
## t = -0.42966, df = 20, p-value = 0.664
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -0.02851065      Inf
## sample estimates:
## mean difference
##      -0.005686012

model_aapl <- lm(log_return ~ spy_return + D_t, data = window_aapl)
model_amzn <- lm(log_return ~ spy_return + D_t, data = window_amzn)
model_googl <- lm(log_return ~ spy_return + D_t, data = window_googl)
model_meta <- lm(log_return ~ spy_return + D_t, data = window_meta)
model_msft <- lm(log_return ~ spy_return + D_t, data = window_msft)
model_nvda <- lm(log_return ~ spy_return + D_t, data = window_nvda)
model_tsla <- lm(log_return ~ spy_return + D_t, data = window_tsla)

summary(model_aapl); summary(model_amzn); summary(model_googl); summary(model_meta);
summary(model_msft); summary(model_nvda); summary(model_tsla)
```

```
##
## Call:
## lm(formula = log_return ~ spy_return + D_t, data = window_aapl)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.021708 -0.005083  0.001339  0.004674  0.023341
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.005755   0.002034  -2.83   0.00725 **
## spy_return   1.533151   0.098570  15.55 < 2e-16 ***
## D_t          0.003958   0.002849   1.39   0.17234
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.009314 on 40 degrees of freedom
## Multiple R-squared:  0.8582, Adjusted R-squared:  0.8511
## F-statistic: 121 on 2 and 40 DF, p-value: < 2.2e-16

##
## Call:
## lm(formula = log_return ~ spy_return + D_t, data = window_amzn)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.042078 -0.007569 -0.000558  0.006371  0.041601
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.006716   0.003064  -2.192   0.0343 *
## spy_return   1.810381   0.148491  12.192 4.74e-15 ***
## D_t          0.004706   0.004291   1.097   0.2794
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01403 on 40 degrees of freedom
## Multiple R-squared:  0.788, Adjusted R-squared:  0.7774
## F-statistic: 74.35 on 2 and 40 DF, p-value: 3.357e-14

##
## Call:
## lm(formula = log_return ~ spy_return + D_t, data = window_googl)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.035565 -0.006778  0.002203  0.009580  0.018703
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```

## (Intercept) -0.0016586  0.0028178  -0.589    0.559
## spy_return   1.4737988  0.1365694  10.792 2.05e-13 ***
## D_t          0.0001059  0.0039467   0.027   0.979
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.0129 on 40 degrees of freedom
## Multiple R-squared:  0.7452, Adjusted R-squared:  0.7325
## F-statistic:  58.5 on 2 and 40 DF,  p-value: 1.328e-12

##
## Call:
## lm(formula = log_return ~ spy_return + D_t, data = window_meta)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.054415 -0.010774 -0.005203  0.014151  0.081128
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003453   0.005452   0.633   0.53
## spy_return   1.635623   0.264248   6.190 2.56e-07 ***
## D_t          0.002961   0.007636   0.388   0.70
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02497 on 40 degrees of freedom
## Multiple R-squared:  0.4892, Adjusted R-squared:  0.4637
## F-statistic: 19.16 on 2 and 40 DF,  p-value: 1.46e-06

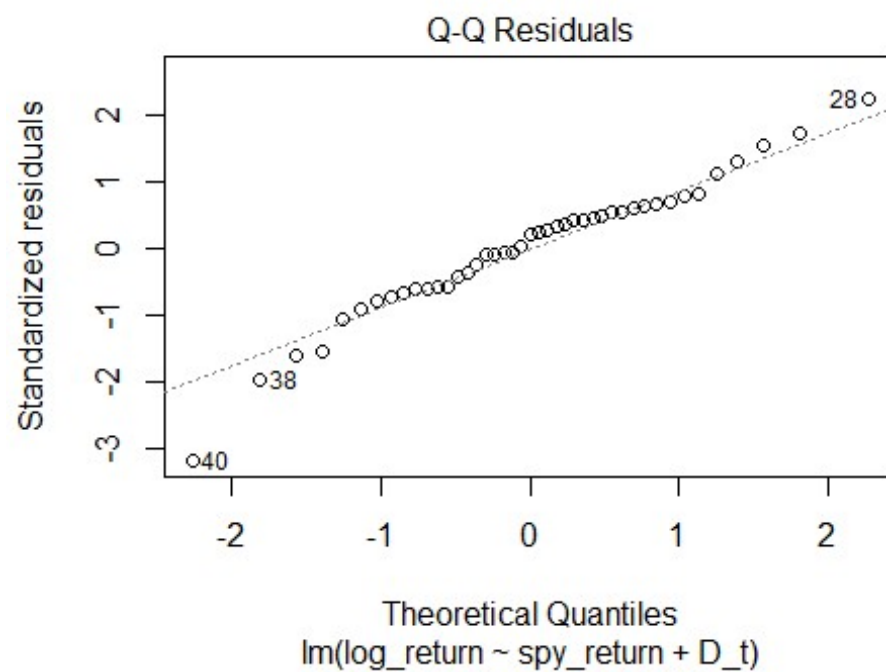
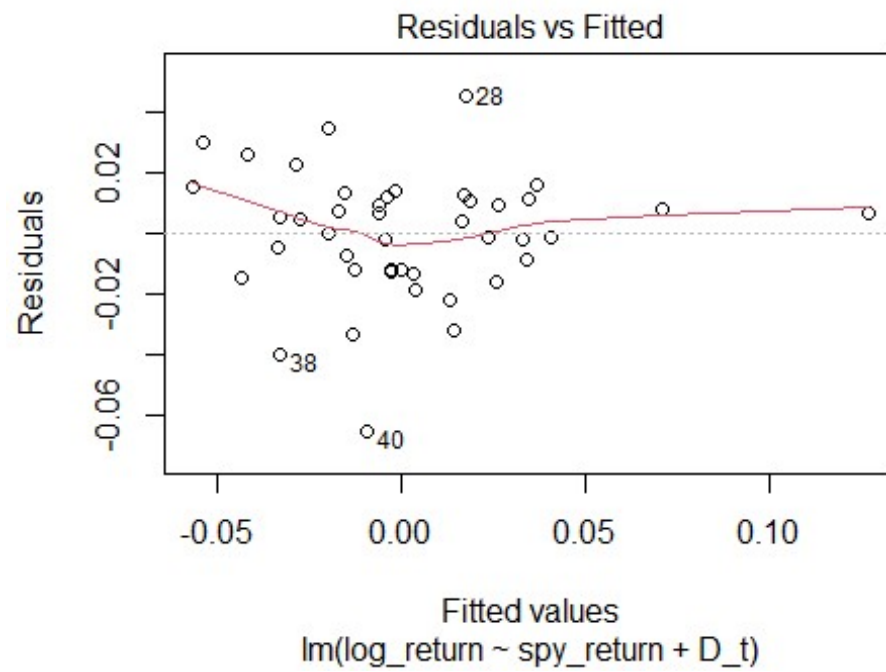
##
## Call:
## lm(formula = log_return ~ spy_return + D_t, data = window_msft)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.0130731 -0.0043174 -0.0004044  0.0043630  0.0147871
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0000955  0.0015959  -0.060   0.953
## spy_return   1.4871481  0.0773473  19.227 <2e-16 ***
## D_t          0.0019147  0.0022352   0.857   0.397
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.007308 on 40 degrees of freedom
## Multiple R-squared:  0.9024, Adjusted R-squared:  0.8975
## F-statistic:  185 on 2 and 40 DF,  p-value: < 2.2e-16

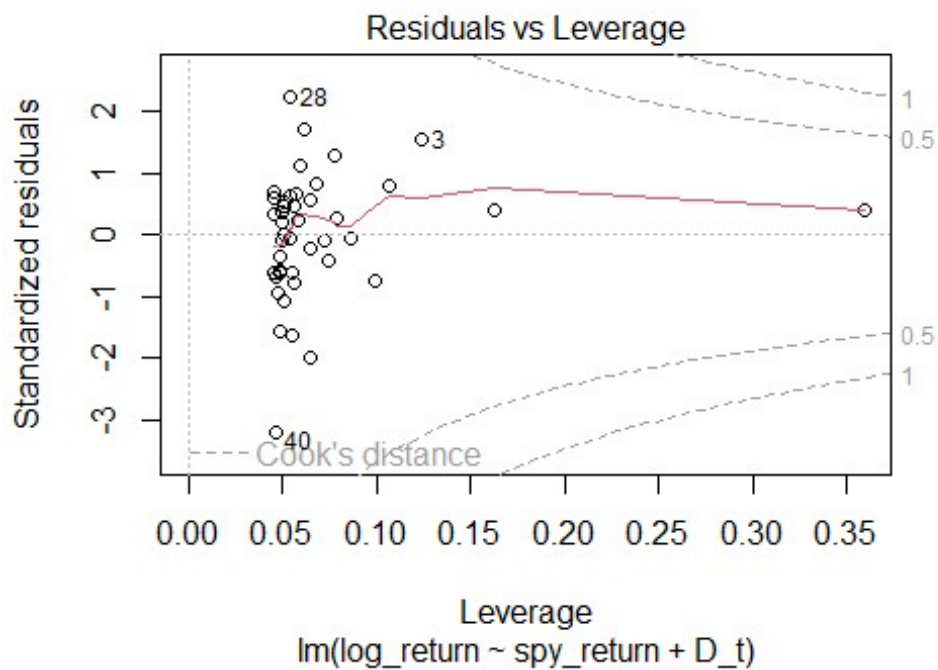
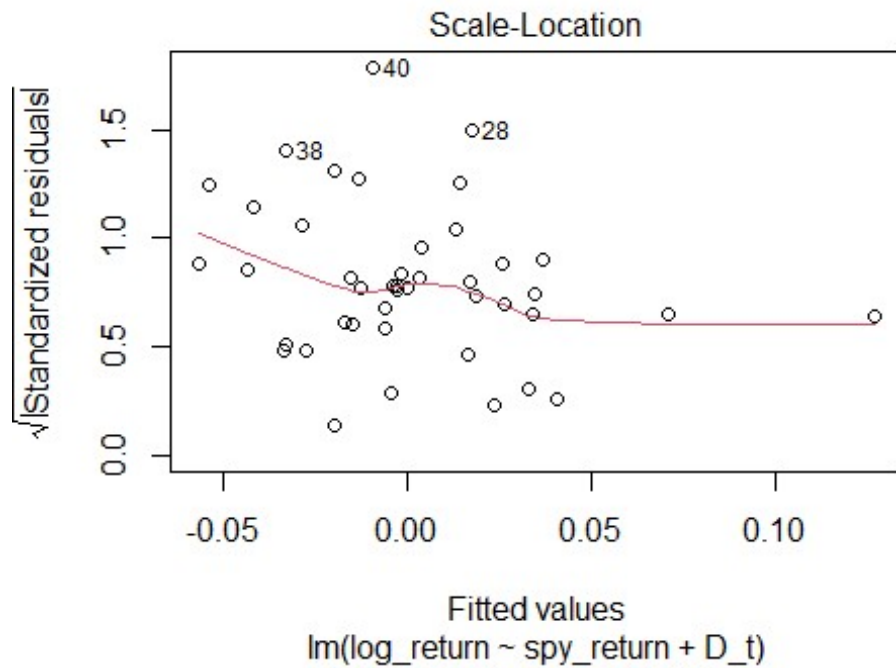
```

```
##
## Call:
## lm(formula = log_return ~ spy_return + D_t, data = window_nvda)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.064833 -0.012098  0.004348  0.011729  0.045205
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.004106   0.004552   0.902   0.373
## spy_return    2.299863   0.220623  10.424 5.75e-13 ***
## D_t          -0.004212   0.006376  -0.661   0.513
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.02085 on 40 degrees of freedom
## Multiple R-squared:  0.7344, Adjusted R-squared:  0.7212
## F-statistic: 55.32 on 2 and 40 DF, p-value: 3.04e-12

##
## Call:
## lm(formula = log_return ~ spy_return + D_t, data = window_tsla)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.099936 -0.020470  0.001189  0.023402  0.078668
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.012208   0.008858  -1.378  0.17580
## spy_return    1.408731   0.429328   3.281  0.00215 **
## D_t          -0.003427   0.012407  -0.276  0.78378
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.04057 on 40 degrees of freedom
## Multiple R-squared:  0.2161, Adjusted R-squared:  0.1769
## F-statistic: 5.513 on 2 and 40 DF, p-value: 0.00768

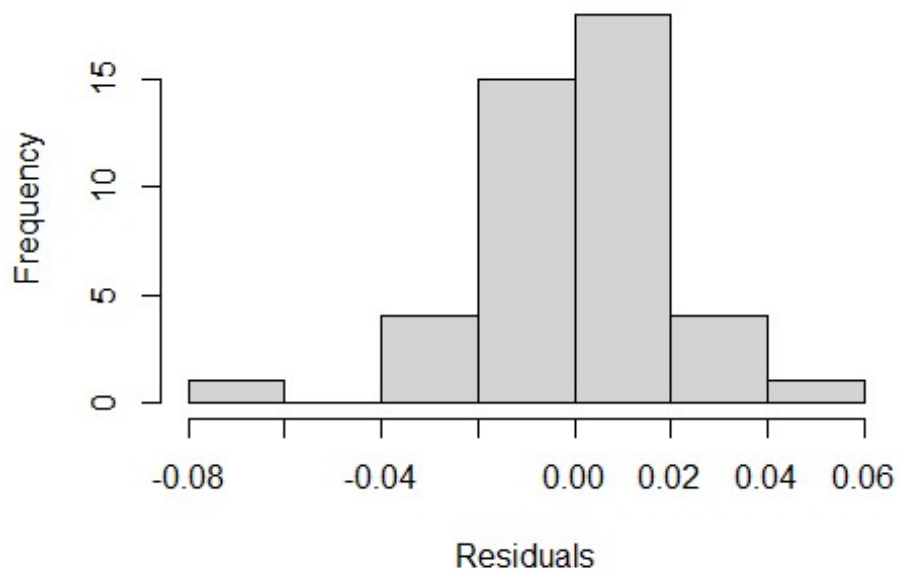
plot(model_nvda)
```





```
hist(residuals(model_nvda),
     main = "Histogram of NVDA Model Residuals",
     xlab = "Residuals")
```

Histogram of NVDA Model Residuals



```
shapiro.test(residuals(model_nvda))
```

```
##  
##  Shapiro-Wilk normality test  
##  
## data:  residuals(model_nvda)  
## W = 0.96436, p-value = 0.1997
```